ELNAZ HEIDARI

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EDUCATION

Ph.D. in Engineering, Clemson University, SC *2022 — 2026*
Dissertation: Explainable physics-guided Transformers for large-scale spatiotemporal flood forecasting.

M.Sc. in Computer Science, Georgia Institute of Technology, GA *2024 — 2026*
Specialization: artificial intelligence and machine learning.

EXPERIENCE

Explainable Transformers for Large-Scale Flood Forecasting **Aug 2022 — Aug 2026**
Graduate Research Assistant, Clemson University *Clemson, SC*

- Built a Temporal Fusion Transformer that forecasts river flow 1–24 h ahead across 15 U.S. regions, **cutting RMSE by 30% vs. NOAA's operational National Water Model**.
- One model learns from 12 weather series and 27 catchment features, replacing per-river models.
- **Generalized zero-shot to 10 unseen regions**, forecasting where no training data exists.
- Added **probabilistic** 10–90% forecast intervals via quantile regression for risk-aware warnings.
- Made forecasts **explainable**: attention and variable-selection weights rank each region's drivers.
- Trained in PyTorch/CUDA on HPC GPU clusters; built pipelines for NetCDF/Zarr climate data.

Deep Recurrent Networks for Flood Event Prediction **Aug 2022 — Dec 2025**
Graduate Research Assistant, Clemson University *Clemson, SC*

- Benchmarked LSTM, GRU, and vanilla RNN models against the National Water Model on flood events in 4 watersheds, using NOAA reanalysis weather and USGS gauge data (*Hydrology*, 2025).
- Quantified uncertainty and compared how each architecture captures flood peaks and timing.

ML Flood Mapping from Social Media and Crowd-Sourced Data **Summer 2023**
Research Assistant, National Water Center (NOAA) *Tuscaloosa, AL*

- Built a decision-tree flood classifier fusing Twitter/X posts and FEMA crowd-sourced reports with NWM inundation maps, yielding more accurate flood-extent maps than the NWM alone.
- Collaborated in a 5-person research team on Python and ArcPy geospatial pipelines, using GitHub pull requests and code review.

Deep Reinforcement Learning and Probabilistic State Estimation **Fall 2026**
Graduate Projects, Georgia Institute of Technology *Atlanta, GA*

- Implemented **PPO from scratch** in PyTorch (no RL libraries), solving continuous-control LunarLander with a **223.7 mean return** (target: 200); ran multi-seed ablations.
- Optimized traffic-signal control as an MDP: model-based planning had **35% fewer congestion violations** than Q-learning and 49% fewer than SARSA.
- Built Kalman and 1,000-particle filters for multi-target tracking and satellite localization.

Generative AI: Diffusion Models, GANs, and VAEs **Spring 2025**
Graduate Project, Georgia Institute of Technology *Atlanta, GA*

- Implemented and benchmarked diffusion models, GANs, and VAEs in PyTorch, analyzing sample quality, diversity, and failure modes such as mode collapse and blur.

Language Modeling and Word Embeddings **Fall 2025**
Graduate Project, Georgia Institute of Technology *Atlanta, GA*

- Built RNN, LSTM, and attention-based language models, **reducing test perplexity by 25%** through architecture and training improvements.
- Trained CBOW and Skip-Gram embeddings on 2M+ tokens, reaching **70% analogy accuracy**.

TECHNICAL SKILLS

Languages	Python, C/C++, Go, R, MATLAB, SQL, Bash
ML Frameworks	PyTorch, TensorFlow, Keras, scikit-learn, CUDA, NumPy, Pandas, SciPy, OpenCV
Deep Learning	Transformers, RNNs/LSTMs, CNNs, diffusion models, GANs, VAEs, attention
ML Methods	Time-series forecasting, uncertainty quantification, transfer learning, explainable AI
Reinforcement Learning	PPO, actor-critic, Q-learning, SARSA, value/policy iteration, MDPs
State Estimation	Kalman filters, particle filters, Bayesian filtering
Data & Infrastructure	HPC GPU clusters, AWS (S3, CLI), NetCDF4, Zarr, HDF5, Xarray, GRIB2, ArcPy, Git

SELECTED GRADUATE COURSES

Reinforcement Learning	Fall 2026	**AI for Robotics**	Fall 2026
Machine Learning	Spring 2026	**AI, Ethics, and Society**	Summer 2026
Natural Language Processing	Fall 2025	**Computer Networks**	Summer 2025
Deep Learning	Spring 2025	**Machine Learning for Trading**	Fall 2024
Data Structures and Algorithms	Spring 2024	**ML for Climate Adaptation**	Fall 2023

PUBLICATIONS

- **Heidari, E.**, and Khan, A. A. (2026). Probabilistic Sub-Daily Streamflow Forecasting in Data-Scarce Catchments Through a Zero-Shot Inductive Transfer Learning Approach. *Journal of the American Water Resources Association (JAWRA).*
- **Heidari, E.**, and Khan, A. A. (2026). Interpretable Hourly Flood Forecasting Leveraging Deep Learning Recurrence and Attention Mechanisms. *Journal of Water Resources Planning and Management* (accepted).
- **Heidari, E.**, Samadi, V., and Khan, A. A. (2025). Leveraging Recurrent Neural Networks for Flood Prediction and Assessment. *Hydrology*, 12(4), 90.
- **Heidari, E.**, Samadi, V., and Popescu, I. (2023). Flood Modeling Across Mountainous and Coastal Watersheds: Coupling Data-Driven and Catchment Physical Attributes. *AGU Fall Meeting.*
- **Heidari, E.**, and Samadi, V. (2023). Application of Deep Recurrent Neural Networks for Flood Prediction and Assessment. *EGU General Assembly*, EGU-16678.